Sokoine University of Agriculture (SUA)
Department of Agricultural Engineering
Electronics and Precision Agriculture Lab (EPAL)

AGE 418
Sensors and Controls in Precision Agriculture

Lecture 5 Compendium:
Proximal Soil Sensors (Part 2)
Nutrients, pH, and Sampling Strategies

Instructor: Dr. Kadeghe Fue
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1 1. The Challenge of Chemical Sensing

In Lecture 4, we discussed sensing *physical* properties (moisture, texture). Today, we tackle the much harder problem: *chemical* sensing (Nutrients N, P, K and pH).

1.1 Why is it hard?

Measuring soil moisture is easy because water has a unique electrical property (high dielectric constant). Measuring Nitrogen (Nitrate ions) is hard because: 1. **Ion Interference:** Many ions (K^+, Na^+, Ca^{2+}) look electrically similar to sensors. 2. **Solid vs Liquid:** Plants eat nutrients dissolved in water, but most soil nutrients are locked in solid particles. Sensors struggle to "see" the solid phase. 3. **Harsh Environment:** Soil is abrasive. Delicate chemical membranes (like those used in lab equipment) break quickly in the field.

The Current Reality

While moisture sensors are cheap and ubiquitous, reliable on-the-go nutrient sensors are still expensive and often require frequent calibration. Therefore, **Soil Sampling** remains the primary method for nutrient mapping in Tanzania.

2 2. Soil Sampling Strategies

Before we deploy high-tech sensors, we must understand how to collect the "Ground Truth" data.

2.1 1. Grid Sampling

The field is divided into equal squares (e.g., 1-hectare grids). Samples are taken from the center of each square.

- **Pros:** Unbiased. Finds variability you didn't know existed. Good for fields with no history.
- **Cons:** Expensive (lab fees). Ignores logical boundaries (e.g., mixing a sandy hill with a clay valley).

2.2 2. Zone Sampling (Smart Sampling)

The field is divided based on existing data layers (Yield maps, EC maps, Topography).

- **Pros:** Cheaper (fewer samples needed). Biologically logical.
- **Cons:** Requires prior data (e.g., an EC map from a Veris machine).

3 3. On-the-Go pH Sensing

Soil pH is the "Master Variable." It controls nutrient availability. If pH is too low (acidic), plants cannot uptake Phosphorus even if you apply fertilizer.

3.1 The Veris MSP (Mobile Sensor Platform)

The most famous commercial machine for this task. It automates the lab process in the field.

1. **Hydraulic Shoe:** A metal shoe drags through the soil to create a furrow.
2. **Sampling:** A scoop grabs a soil sample every few seconds.
3. **Measurement:** The soil is pressed against an **Ion Selective Electrode (ISE)** made of antimony.
4. **Cleaning:** Water jets wash the sensor before the next sample.

Speed: Maps roughly 5-10 hectares per hour.

4 4. Optical Soil Sensing (Vis-NIR)

Visible and Near-Infrared (Vis-NIR) spectroscopy is a non-destructive method to estimate soil properties.

4.1 Principle

Light is shined onto the soil (using a probe that cuts a furrow). The soil absorbs certain wavelengths based on its chemical bonds.

- **Carbon (Organic Matter):** Darker soils absorb more light. Strong absorption in NIR bands.
- **Moisture:** Water has distinct absorption bands at 1450nm and 1950nm.
- **Total Nitrogen:** Often correlates with Carbon, so Vis-NIR estimates Total N well.

4.2 Limitations

Vis-NIR is great for Carbon and Moisture but **poor for Phosphorus and Potassium**, as these minerals do not have strong spectral features in the NIR range.

5 5. Electrochemical Sensors (ISE)

These are "Lab-on-a-Chip" technologies.

5.1 Ion Selective Electrodes (ISE)

An electrode with a membrane that only allows specific ions (e.g., Nitrate NO_3^- or Potassium K^+) to pass through, creating a voltage potential.

- **Pros:** Direct measurement of the nutrient ion.
- **Cons:** The "Membrane Problem." Soil particles scratch the membrane. Calibration drifts quickly (every hour).
- **Status:** Mostly used in slurry tankers (manure sensing) or hydroponics, rather than direct soil dragging.

6 6. Gamma Ray Spectroscopy

This is a **Passive** sensor technology used for mapping soil texture and some nutrients.

6.1 The Physics

All soil emits low-level natural radiation from radioisotopes.

1. **Potassium-40 (^{40}K):** Directly correlates to total soil Potassium.
2. **Thorium (^{232}Th) & Uranium (^{238}U):** These elements bind to clay particles.

6.2 The "Soil Sensor" (e.g., Gammatrics)

A sensor mounted on a tractor simply "listens" to the radiation.

- **High Radiation Count:** Indicates Clay soil (and usually high Potassium).
- **Low Radiation Count:** Indicates Sandy soil.

Benefit: It is non-contact, unaffected by moisture, and creates excellent texture maps.

7 7. Variable Rate Application (VRA)

The ultimate goal of sensing is to act.

7.1 VRA Lime

Using the pH map generated by the Veris machine, we create a prescription.

- **Acidic Zones (pH < 5.5):** Apply high rate of Lime.
- **Neutral Zones (pH 6.5):** Apply zero Lime.

Impact: Massive cost savings and better nutrient availability.

7.2 VRA Fertilizer

Using Zone Sampling data (N, P, K), we apply fertilizer only where levels are below critical thresholds.

8 8. Lecture Assessment

8.1 Part A: Standard Concept Questions

1. Why is chemical sensing of soil (Nutrients) more difficult than physical sensing (Moisture)?
2. Explain the difference between **Grid Sampling** and **Zone Sampling**. Which one is more cost-effective for a farm with high variability?
3. What is the "Master Variable" in soil chemistry, and why is it critical to map it before applying fertilizer?
4. Describe the working principle of the **Veris MSP** on-the-go pH sensor.
5. What is **Vis-NIR Spectroscopy**? Which soil properties can it estimate accurately, and which ones does it struggle with?
6. Explain how **Gamma Ray Spectroscopy** works. What specific nutrient can it measure directly?
7. What is an **Ion Selective Electrode (ISE)**? What is the main engineering challenge in using ISEs for direct soil sensing?
8. Define **Variable Rate Application (VRA)**. How does a VRA Lime map differ from a traditional blanket application?
9. Why do we say that Electrical Conductivity (EC) maps are often used as a "template" for Zone Sampling?
10. What is the relationship between Soil Organic Matter (OM) and soil color, and how do optical sensors exploit this?

8.2 Part B: Advanced Analysis (Open Source Data)

These questions emphasize the use of open-source datasets available for Africa.

1. **AfSIS / SoilGrids Analysis:** Visit **SoilGrids.org** (part of the Africa Soil Information Service). Navigate to the **Kilimanjaro Region**. *Task:* View the "Soil pH in Water" layer. Identify areas with acidic soils (Red/Orange). Compare this with the "SOC" (Soil Organic Carbon) layer. Is there a visual correlation? How would you use this free data to plan a physical sampling mission?
2. **Sampling Strategy Design:** Using **Google Earth**, locate a 50-hectare block of farmland in **Dakawa, Morogoro**. *Task:* Design a "Zone Sampling" plan. Use visual cues from the satellite image (e.g., darker soil areas vs. lighter sandy areas, topography) to draw 3 distinct management zones. Explain why you drew the boundaries where you did.
3. **Digital Soil Mapping (DSM) Logic:** DSM uses "Covariates" to predict soil properties. *Task:* Explain how you would use a **Digital Elevation Model (DEM)** (Topography) and **Sentinel-2 NDVI** (Vegetation health) as proxies to predict where Organic Matter might be highest in a field. (Hint: Water and biomass accumulate in low spots).

4. **Gamma Ray Interpretation:** You are given a Gamma Ray map of a field in Iringa. The map shows very high counts of Thorium. *Task:* Based on the lecture, what soil texture does this likely indicate? If you were planting Maize (which prefers loam) vs. Rice (which tolerates clay), how would this map influence your crop placement?

5. **Economic Case for VRA Lime:** A farmer in Mbeya has a 100-acre field. Soil tests show 40% of the field is acidic (pH 5.0) requiring 2 tons/acre of lime. The other 60% is neutral (pH 6.5) requiring 0 lime. *Task:* Calculate the cost difference between a "Blanket Application" (2 tons/acre everywhere) vs. "Variable Rate Application." Assume Lime costs TZS 150,000 per ton. Does the savings justify the TZS 2,000,000 cost of hiring a mapping service? *Note:* Blanket Cost = 100 acres * 2 tons * 150k. VRA Cost = (40 acres * 2 tons * 150k) + 2M Service Fee.